

Geometry: Two-Column Proofs — Kite & Trapezoid



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Name: _____

Date: _____

Score: / 3 proofs

DIRECTIONS

Fill in the BLANK statements or reasons to complete each two-column proof. Use proper notation and geometric theorems.

1 Proof 1 | Kite: Proving Non-Vortex Angles Are Congruent

Given: Kite $ABCD$ with $\overline{AB} \cong \overline{AD}$ and $\overline{CB} \cong \overline{CD}$

Prove: $\angle B \cong \angle D$

Statements	Reasons
1. $\overline{AB} \cong \overline{AD}, \overline{CB} \cong \overline{CD}$	1. Given
2. $\overline{AC} \cong \overline{AC}$	2. _____
3. $\triangle ABC \cong \triangle ADC$	3. SSS Congruence Postulate
4. _____	4. CPCTC

2 Proof 2 | Kite: Proving the Diagonals Are Perpendicular

Given: Kite $ABCD$; $\overline{AB} \cong \overline{AD}$, $\overline{CB} \cong \overline{CD}$; diagonals intersect at E

Prove: $AC \perp BD$

Statements	Reasons
1. $\overline{AB} \cong \overline{AD}, \overline{CB} \cong \overline{CD}$	1. Given
2. $\overline{AE} \cong \overline{AE}$	2. _____
3. _____	3. SSS Congruence Postulate
4. $\angle AEB \cong \angle AED$	4. CPCTC
5. $AC \perp BD$	5. _____



3 Proof 3 | Isosceles Trapezoid: Proving the Diagonals Are Congruent

Given: Isosceles trapezoid $ABCD$; $AB \parallel CD$; $\overline{AD} \cong \overline{BC}$; $\angle DAB \cong \angle CBA$

Prove: $\overline{AC} \cong \overline{BD}$

Statements	Reasons
1. $\overline{AD} \cong \overline{BC}$, $\angle DAB \cong \angle CBA$	1. Given
2. $\overline{AB} \cong \overline{AB}$	2. _____
3. $\triangle DAB \cong \triangle CBA$	3. _____
4. $\overline{AC} \cong \overline{BD}$	4. CPCTC



Answer Key & Solutions

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TEACHER NOTES

Blanks: Proof 1 — Step 2 reason: Reflexive Property; Step 4 statement: angle B = angle D (CPCTC). Proof 2 — Step 2 reason: Reflexive; Step 3 statement: triangles ABE = ADE (SSS); Step 5 reason: congruent supplementary angles. Proof 3 — Steps 2 & 3 reasons: Reflexive Property; SAS Congruence Postulate.

1 Proof 1 | Kite: Proving Non-Vortex Angles Are Congruent

Given: Kite $ABCD$ with $\overline{AB} \cong \overline{AD}$ and $\overline{CB} \cong \overline{CD}$

Prove: $\angle B \cong \angle D$

Statements	Reasons
1. $\overline{AB} \cong \overline{AD}, \overline{CB} \cong \overline{CD}$	1. Given
2. $\overline{AC} \cong \overline{AC}$	2. Reflexive Property
3. $\triangle ABC \cong \triangle ADC$	3. SSS Congruence Postulate
4. $\angle B \cong \angle D$	4. CPCTC

2 Proof 2 | Kite: Proving the Diagonals Are Perpendicular

Given: Kite $ABCD$; $\overline{AB} \cong \overline{AD}$, $\overline{CB} \cong \overline{CD}$; diagonals intersect at E

Prove: $AC \perp BD$

Statements	Reasons
1. $\overline{AB} \cong \overline{AD}, \overline{CB} \cong \overline{CD}$	1. Given
2. $\overline{AE} \cong \overline{AE}$	2. Reflexive Property
3. $\triangle ABE \cong \triangle ADE$	3. SSS Congruence Postulate
4. $\angle AEB \cong \angle AED$	4. CPCTC
5. $AC \perp BD$	5. Congruent supplementary angles form right angles



3 Proof 3 | Isosceles Trapezoid: Proving the Diagonals Are Congruent

Given: Isosceles trapezoid $ABCD$; $AB \parallel CD$; $\overline{AD} \cong \overline{BC}$; $\angle DAB \cong \angle CBA$

Prove: $\overline{AC} \cong \overline{BD}$

Statements	Reasons
1. $\overline{AD} \cong \overline{BC}$, $\angle DAB \cong \angle CBA$	1. Given
2. $\overline{AB} \cong \overline{AB}$	2. Reflexive Property
3. $\triangle DAB \cong \triangle CBA$	3. SAS Congruence Postulate
4. $\overline{AC} \cong \overline{BD}$	4. CPCTC

